

Vehicle Parameter Identification — Report Outline

1 Vehicle Parameter Identification

The dynamic bicycle model used by the MPC controller requires physical parameters that cannot be taken from datasheets alone. This section describes the experimental procedures used to identify them on the real F1/10th vehicle.

All tests use a common ROS 2 framework that records time-stamped sensor data at approximately 200 Hz: VESC odometry (speed from ERPM), VESC IMU (3-axis accelerometer and gyroscope), and motor state (current, voltage, temperature). Each test saves raw data to CSV for post-processing and automatically updates a central `vehicle_params.yaml` file.

1.1 Dynamic Bicycle Model

The MPC uses a 7-state dynamic bicycle model with state $\mathbf{x} = [x, y, \psi, v_x, v_y, \omega, \omega_w]$ and control inputs $\mathbf{u} = [\delta, T_{\text{motor}}]$. The equations of motion are:

$$\dot{v}_x = \frac{1}{m} (F_x - F_{yf} \sin \delta + m v_y \omega) \quad (1)$$

$$\dot{v}_y = \frac{1}{m} (F_{yf} \cos \delta + F_{yr} - m v_x \omega) \quad (2)$$

$$\dot{\omega} = \frac{1}{I_z} (l_f F_{yf} \cos \delta - l_r F_{yr}) \quad (3)$$

$$\dot{\omega}_w = \frac{1}{I_w} \left(\frac{T_{\text{motor}}}{G} - F_x R_w \right) \quad (4)$$

where G is the gear ratio, R_w the wheel radius, and I_w the drivetrain inertia. Lateral tire forces use a linear model:

$$F_{yf} = C_{\alpha f} \alpha_f, \quad F_{yr} = C_{\alpha r} \alpha_r \quad (5)$$

with slip angles

$$\alpha_f = \delta - \arctan \frac{v_y + l_f \omega}{v_x}, \quad \alpha_r = -\arctan \frac{v_y - l_r \omega}{v_x}. \quad (6)$$

The longitudinal tire force is $F_x = C_x \kappa$, where κ is the slip ratio. Table 1 lists all model parameters and the test used to identify each one.

1.2 Tire Friction Coefficient

Theory. At steady-state circular driving the centripetal acceleration is $a_y = v^2/R$. The maximum sustainable a_y before the tires begin to slide gives the friction coefficient:

$$\mu = \frac{a_{y,\text{max}}}{g}. \quad (7)$$

The IMU measures a_y directly, independent of odometry errors.

Procedure. The car drives circles at a fixed steering angle with speed increasing in 0.5 m s^{-1} steps from 1.0 m s^{-1} to 4.0 m s^{-1} . At each speed, 8 s of steady-state IMU data is recorded.

Data. [Figure not yet generated: friction·ay·vs·speed.pdf]

Table: speed | $\bar{a}_y \pm \sigma$ | $\mu = a_y/g$.

CSV columns used: imu_ay, speed_setpoint, phase. Group by speed_setpoint, compute mean and std of imu_ay in the recording phase. Before aggregation, use bias-corrected IMU channels and keep only near steady-state samples where $|a_y| \approx |v_x \omega|$.

Uncertainty.

- IMU bias: 2 s stationary calibration; residual $< 0.05 \text{ m s}^{-2}$.
- Surface-dependent: result only valid for the tested floor.
- Temperature-dependent: rubber grip varies with tire temperature.

1.3 Cornering Stiffness

Theory. In steady-state cornering ($\dot{v}_y = \dot{\omega} = 0$), the front and rear lateral forces and slip angles reduce to:

$$F_{yf} = m a_y \frac{l_r}{L}, \quad F_{yr} = m a_y \frac{l_f}{L}, \quad (8)$$

$$\alpha_f \approx \delta - \frac{l_f \omega}{v_x}, \quad \alpha_r \approx \frac{l_r \omega}{v_x}, \quad (9)$$

where ω is the IMU yaw rate and v_x the ERPM-based speed. The cornering stiffnesses follow from $C_{\alpha f} = F_{yf}/\alpha_f$ and $C_{\alpha r} = F_{yr}/\alpha_r$. The understeer gradient is

$$K_{us} = \frac{m g}{L} \left(\frac{l_f}{C_{\alpha f}} - \frac{l_r}{C_{\alpha r}} \right). \quad (10)$$

Procedure. Same steady-state circles as the friction test, at speeds from 1.0 m s^{-1} to 3.5 m s^{-1} . At each speed, 5 s of v_x , ω (IMU), and a_y (IMU) are recorded.

Data. [Figure not yet generated: cornering·Fy·vs·alpha.pdf]

[Figure not yet generated: cornering·alpha·vs·speed.pdf]

Table: speed | α_f | α_r | $C_{\alpha f}$ | $C_{\alpha r}$.

CSV columns used: odom_vx, imu_gz, imu_ay, cmd_steering, phase. Compute α_f , α_r from (9), then F_{yf} , F_{yr} from (8). Bias-corrected IMU data is used, and samples with large $|a_y - v_x \omega|$ consistency error are filtered out.

Uncertainty.

- Relies on measured m , l_f , l_r ($\pm 1\text{--}2 \text{ mm}$).
- Small-angle approximation breaks down above $\sim 5^\circ$ slip.
- IMU yaw-rate bias drift (mitigated by 2 s calibration).
- ERPM noise affects v_x and thus slip-angle calculation.

1.4 Longitudinal Tire Stiffness

Theory. The slip ratio is

$$\kappa = \frac{v_{\text{wheel}} - v_{\text{body}}}{\max(|v_{\text{wheel}}|, |v_{\text{body}}|)}, \quad (11)$$

where v_{wheel} comes from ERPM and v_{body} from IMU integration. In the linear region ($|\kappa| < 0.15$):

$$F_x = C_x \kappa, \quad F_x = m a_x. \quad (12)$$

Procedure. Accelerate from standstill to 3 m s^{-1} , then brake to stop. Both v_{wheel} and v_{body} are recorded simultaneously. A 2 s stationary bias calibration precedes the test.

Data. [Figure not yet generated: long`v`comparison.pdf]

[Figure not yet generated: long`Fx`vs`kappa.pdf]

CSV columns used: `odom_vx` (v_{wheel}), `v_imu` (v_{body}), `imu_ax`, `slip_ratio`, `phase`. Use bias-corrected a_x for both force estimation and IMU velocity integration.

Uncertainty.

- IMU integration drift accumulates; test kept short ($< 10 \text{ s}$).
- 4WD distributes torque through differentials; C_x is an aggregate combined value.
- Rolling resistance and aerodynamic drag contribute to $F_x = m a_x$ (small at low speed).

1.5 Motor Torque Mapping

Theory. Two independent estimates of wheel force:

$$F_{\text{current}} = K_t I_{\text{motor}} \frac{G}{r_{\text{eff}}}, \quad (13)$$

$$F_{\text{accel}} = m a_x \quad (\text{IMU}), \quad (14)$$

where $K_t = 60/(2\pi K_v)$ and $G = 11.82$ is the gear ratio. Plotting F_{accel} against I_{motor} yields the effective current-to-force gain (slope) and drivetrain losses (intercept).

Procedure. Accelerate to target speeds (1.5, 2.0, 3.0, 4.0 m s^{-1}), recording motor current and IMU a_x simultaneously. Braking (regenerative) current is measured when the car decelerates.

Data. [Figure not yet generated: torque`I`vs`F.pdf]

[Figure not yet generated: torque`current`vs`time.pdf]

Table: speed | $\bar{I}$ | peak I | F_{net} | efficiency.

CSV columns used: `motor_current`, `imu_ax`, `cmd_speed`, `phase`. Acceleration is computed from bias-corrected IMU a_x .

Uncertainty.

- Motor K_v from datasheet ($\pm 5\%$ typical for hobby motors).
- r_{eff} measurement accuracy ($\pm 1 \text{ mm} \Rightarrow \sim 2\%$ error on force).
- Gear efficiency not measured separately; absorbed into effective K_t .

1.6 Maximum Dynamics

Theory. Direct measurement of the vehicle's performance envelope: v_{max} , a_{max} (peak acceleration), and a_{brake} (peak deceleration). These define the MPC input/state constraints.

Procedure. Command maximum speed for 5 s (acceleration phase), then command zero (braking phase). Both ERPM-based speed and IMU a_x are recorded. A velocity plateau detector checks whether the car actually reached top speed.

Data. [Figure not yet generated: max'dyn'speed'vs'time.pdf]

CSV columns used: `odom.vx`, `imu.ax`, `phase`. Both peak acceleration/deceleration and IMU-integrated speed use bias-corrected IMU a_x .

Uncertainty.

- Odom speed unreliable during braking (wheel lock/slip); use IMU.
- IMU-integrated velocity drifts; odom more reliable during steady acceleration.
- Battery voltage sag under load reduces effective $v_{\max}$.

1.7 Steering Actuator Rate

Theory. The servo has a finite angular velocity. By commanding step changes $0 \rightarrow \delta_{\max}$ and observing the 10%–90% rise time of the IMU yaw rate:

$$\dot{\delta}_{\max} = \frac{\Delta\delta}{\Delta t_{10-90}}. \quad (15)$$

Procedure. Drive at constant 1.5 m s^{-1} . Alternate steering between $+\delta$ and $-\delta$ for 6 transitions. Record IMU yaw rate ω_z and commanded steering angle.

Data. [Figure not yet generated: steering'step'response.pdf]

Table: transition | $\Delta\delta$ | Δt | rate (rad s^{-1}).

CSV columns used: `imu.gz`, `cmd.steering`, `phase`.

Uncertainty.

- Yaw rate includes vehicle dynamics response, not only servo motion.
- At low speed (1.5 m s^{-1}) the dynamic lag is small relative to the servo speed.
- Servo speed may depend on battery voltage.

1.8 Parameter Summary

Table 1 compares tested values with the F1/10th gym simulation defaults.

Table 1: Identified vehicle model parameters.

Parameter	Symbol	Measured	Sim default	Unit
Friction coeff.	μ	—	1.049	—
Front corn. stiff.	$C_{\alpha f}$	—	4.718	N rad^{-1}
Rear corn. stiff.	$C_{\alpha r}$	—	5.456	N rad^{-1}
Long. stiffness	C_x	—	300	N
Max drive torque	$T_{\max}$	—	22.9	N m
Max brake torque	$T_{\min}$	—	-24.1	N m
Max velocity	$v_{\max}$	—	20.0	m s^{-1}
Max acceleration	$a_{\max}$	—	9.51	m s^{-2}
Max deceleration	a_{brake}	—	10.0	m s^{-2}
Max steer. rate	$\dot{\delta}_{\max}$	—	3.2	rad s^{-1}